
When the reference hides uncertainty: a data-based simulation audit

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Abstract

1 Scoring the same nominal 95% intervals against a sampled reference gives 98.9–
2 99.6% inclusion, versus 92.3–93.6% coverage of the generating-pool effect, in a
3 single-cell simulation audit at 800 cells. An exact variance calculation explains the
4 discrepancy: under a halving of expression, sharing the sampled reference mean
5 removes three quarters of the reference-arm variance contribution from the error
6 being assessed. We trace the audit to a 50-cell reporting rule previously justified
7 by point-estimate recovery. In 5,000 studies per setting, the original interval
8 rejects a true null in 9.8–32.2% of studies; a Welch control without bootstrap
9 resampling noise still rejects in 8.9–30.8%. A further 32,000-study comparison
10 evaluates an interior point of the flag-wide band, larger bootstrap budgets, BCa and
11 bootstrap- t . None of the tested methods satisfies coverage, detection and nominal
12 null calibration jointly at 30 or 50 cells. A short external-control illustration
13 distinguishes population alignment from numerical agreement. The evidence is
14 conditional on empirical cell pools: it identifies a failed reporting claim and tests
15 concrete repairs, without validating a replacement threshold or a patient world
16 model.

17 1 Motivation: evaluate the reported uncertainty

18 A nominal 95% interval can appear to cover its reference almost perfectly while covering the intended
19 parameter less often. In our single-cell simulation audit, changing only the scoring reference changes
20 inclusion from 98.9–99.6% to 92.3–93.6% at 800 cells. The source of the discrepancy is explicit: the
21 estimate and reference share a sampled mean, canceling part of the error that the interval is meant to
22 cover. We derive that variance difference and examine its consequences for a reporting rule.

23 The case comes from our unpublished colorectal project, where point-estimate recovery had been
24 used to justify a 50-cell reporting boundary. We audit that recorded inference using simulations
25 resampled from observed single-cell distributions. The analysis covers GUCA2A, one labeling axis
26 and one weighting, with ten SMC and six KUL3 patients. Each study draws cells from a two-patient
27 pool; repeated draws quantify performance conditional on those source pools, not uncertainty over
28 new patients. The contribution is this documented failure, its quantitative explanation and paired
29 interval comparisons.

30 Section 2 assesses the original interval, a resampling-free control and skew-aware alternatives at
31 specified counts, including an interior point of the flag-wide band. A short synthetic external-control
32 illustration (Section 3) separates a related question: whether the reference names the intended
33 population. It supports the interpretation of the audit rather than providing a second empirical case.

34 **Relation to existing validation work.** Simulation methodology already separates estimands, meth-
35 ods and performance measures [Burton et al., 2006, Morris et al., 2019]; ICH E9(R1) specifies

the population targeted by the clinical question [International Council for Harmonisation, 2019]. Generator–analysis alignment is studied through congeniality [Meng, 1994], semi-synthetic benchmarks [Franklin et al., 2014, Curth et al., 2021], interventional evaluation [Gentzel et al., 2019], and the inverse crime [Wirgin, 2004, Kaipio and Somersalo, 2007]. Generators can also disadvantage good estimators [Shaw et al., 2026]. Simulation-based calibration checks Bayesian computation using prior-predictive ranks [Talts et al., 2018], with sensitivity depending on the test quantities [Modrák et al., 2025]. Our frequentist assessment conditions on empirical pools and specified effects; it applies established validation principles to the claim made for this artifact.

Connection to patient simulation. Digital Twin Generators learn conditional clinical trajectories [Alam et al., 2024]; ClinicalGAN evaluates personalized sequence generation and trial-dropout prediction [Chandra et al., 2024]. Those modeling tasks differ from assessing the coverage of a downstream treatment-effect interval. Our audit addresses the latter validation layer, without evaluating either system. A supporting learned-generator check shows why generator complexity does not break an estimator–reference identity (Appendix C). Clinical transfer additionally requires source comparability, as in held-out external-control validation [Ventz et al., 2019], and an explicit causal design [Hernán and Robins, 2016, Schuler et al., 2022]. These are requirements for applying the audit’s lessons to patient world models, not results established by this study.

2 Audit of a cell-count reporting rule

Application and reporting policy. The biological application separates two explanations for differentiation-marker loss in colorectal cancer: fewer marker-producing cells (a *compositional* change), or lower output from the cells that remain (an *intrinsic* change). These explanations suggest different biological interpretations. An empirical comparison of per-cell means requires the relevant population to be observed in both tissues. Let f_N, f_T denote the relevant cell fractions in reference and diseased tissue, and m_N, m_T their mean per-cell output. A reference-based decomposition, related to demographic standardization [Kitagawa, 1955], is

$$\underbrace{(f_T - f_N)m_N}_{\text{compositional}} + \underbrace{f_N(m_T - m_N)}_{\text{intrinsic}} + \underbrace{(f_T - f_N)(m_T - m_N)}_{\text{interaction}} = f_T m_T - f_N m_N. \quad (1)$$

The interaction is reported separately. If no relevant cells are observed in an arm, its sample mean is undefined; the identity cannot supply the missing mean. In a synthetic population with no such cells, the corresponding conditional mean itself has no referent. Substituting zero in either case turns an unavailable contrast into an apparent effect.

The implemented reporting rule returns an estimate at $n \geq 50$, flags a wide interval at $20 \leq n < 50$, and returns NOT ESTIMABLE below 20, where n is the relevant-cell count in diseased tissue. Unavailable reported contrasts are stored as None, not zero. The diagnostic oracle calculations examined below also retain raw arithmetic before this reporting gate. We evaluate the boundary directly and retain pre-gate intervals at smaller counts to diagnose the curve’s behavior.

The claim being re-evaluated. An internal assessment (Artifact A) treated the 50-cell boundary as validated because recovery ratios were 0.85–1.08 above it and 1.18 in the 20–49-cell band, using six draws per setting at 1,200 cells per arm. The design record (Artifact B) instead required at least 90% interval coverage and 80% exclusion of zero at a halving of expression. Appendix F and the anonymized supporting record make this mismatch inspectable.

2.1 Simulation and inferential targets

The design document fixed a detectable halving of per-cell output ($s = 0.5$), coverage of the target in at least 90% of defined outputs, and discrimination of at least 80%, defined as the probability that the interval excludes zero at that effect. The implementation scored coverage against a draw-dependent reference benchmark; we retain that score and add coverage against the generating pool mean. The candidate *ok* is the first evaluated count or bin meeting both criteria; *wide* is the first meeting coverage alone. The committed values are 50 and 20. The first-crossing definition does not require larger counts to qualify.

83 The calibration uses paired subsets of the published colorectal atlas [Lee et al., 2020]: ten patients from
84 GSE132465 (SMC) and six from GSE144735 (KUL3). We analyze GUCA2A in the differentiated
85 population under fixed lineage labeling. Each draw samples, with replacement, 2,000 cells per arm
86 from an eligible two-patient pool; the reference mature fraction is $f_N = 0.40$. The implemented pool
87 combines cells from both tissues before drawing either synthetic arm. Because the design document
88 did not specify tissue eligibility, we also evaluate a reference-tissue-only pool. Both arms sample the
89 same eligible pool, with an imposed expression shift. A nominal 95% percentile interval resamples
90 mature cells in both arms, with fractions held fixed and 200 inner draws unless stated otherwise.
91 Undefined intervals are excluded from rate denominators and reported as abstentions (Appendix B).

92 The stored “parametric” intrinsic reference is

$$i_{\text{req}}(D_N; s) = f_N(s - 1)\bar{Y}_N, \quad (2)$$

93 where $\bar{Y}_N$ is the mature-cell mean of the *drawn* reference arm. Let P be the eligible two-patient
94 source pool, M_P its mature-cell count, and Y_j its unshifted target-gene counts. The population
95 parameter of this finite empirical distribution is instead

$$\mu_P = \frac{1}{M_P} \sum_{j \in P, \text{mature}} Y_j, \quad \theta_P(s) = f_N(s - 1)\mu_P. \quad (3)$$

96 We compute μ_P from all eligible source cells before either synthetic arm is sampled. Given P , the
97 generator has $E[\bar{Y}_N] = \mu_P$ and $E[\bar{Y}_T] = s\mu_P$, making Equation 3 its true intrinsic mean contrast.
98 It is fixed across cell draws from that pool. Patient-pair selection changes P ; reported coverage
99 averages this conditional-pool question over the selected pairs. An all-patient pooled mean would
100 use a different population from the generator’s two-patient draw. Both targets are conditional on the
101 source-pool design stated above.

102 The historical grid and precision follow-up separate selection from direct evaluation; the additional
103 repair comparison is described in Section 2.5. The controlled eight-seed grid retains inclusion of
104 Equation 2. An initial 200-replicate diagnostic reproduced its first-seed results and added fixed-pool
105 coverage and null rejection. The precision follow-up uses 5,000 fresh studies per cohort, pool, count
106 (5, 50, 100, 800) and shift ($s = 0.5, 1$), split across five seed streams: 160,000 studies in total. It
107 samples the exact marginal law of mature-cell counts in the original generator; non-mature cells
108 do not enter this contrast. On each study we compare the original 200-resample percentile interval
109 with a Welch–Satterthwaite interval [NIST/SEMATECH]. This unequal-variance t approximation
110 provides an interval with no bootstrap resampling noise; it is not chosen for robustness to skewness.
111 All 320,000 intervals are finite. The largest binomial Monte Carlo standard error is 0.71 percentage
112 points, conditional on the empirical source pools. Appendix A specifies the sampling and interval
113 calculations.

114 2.2 Fixed-pool coverage and false positives

115 Two standards answer different questions (Table 1). At $s = 0.5$, percentile coverage is below the
116 historical minimum of 90% in three conditions; SMC/reference reaches 91.0% [90.2, 91.8]. All four,
117 including SMC/reference, remain below the intervals’ nominal 95% level, with every Wilson upper
118 endpoint below 95%. The historical tolerance does not establish nominal calibration.

119 At $s = 1$, the fixed-pool target is zero, so null rejection is exactly one minus coverage among
120 defined intervals. The reported 9.8–32.2% null rejection is therefore 67.8–90.2% null coverage,
121 rather than an independent calibration criterion. Coverage is lower at the null than at the halving in
122 all four conditions. Detection at $s = 0.5$ is a distinct quantity: exclusion of zero under a nonzero
123 effect, here 47.5–76.2%. All detection upper endpoints are below the historical 80% requirement; all
124 null-rejection lower endpoints exceed nominal 5%. Wilson intervals quantify Monte Carlo precision
125 conditional on the source pools.

126 2.3 Shared-reference scoring removes part of the estimation error

127 At 800 cells, evaluating the percentile intervals against the sampled reference gives 98.9–99.6%
128 inclusion, compared with 92.3–93.6% coverage of the fixed-pool effect. The paired differences are
129 5.8–6.8 percentage points with Monte Carlo standard errors of 0.33–0.36 points. Only the scoring

Table 1: At the 50-cell reporting boundary, substituting Welch intervals improves coverage but does not restore the required performance. Rates (%) and 95% Wilson Monte Carlo intervals use 5,000 studies per cohort, pool and effect. Methods share the same samples. Coverage targets the fixed-pool effect at $s = 0.5$; null rejection uses $s = 1$; detection uses $s = 0.5$. Historical requirements are at least 90% alternative coverage and 80% detection. Nominal coverage is 95%; null rejection is one minus null coverage and should be 5%.

Cohort / pool	Interval	Coverage	Null rejection	Detection
SMC / pooled	Percentile	87.1 [86.2, 88.0]	25.4 [24.2, 26.6]	51.1 [49.7, 52.4]
SMC / pooled	Welch	90.2 [89.3, 91.0]	23.5 [22.3, 24.7]	50.7 [49.3, 52.1]
SMC / reference	Percentile	91.0 [90.2, 91.8]	9.8 [9.0, 10.7]	76.2 [75.0, 77.3]
SMC / reference	Welch	92.5 [91.8, 93.2]	8.9 [8.1, 9.7]	77.6 [76.5, 78.8]
KUL3 / pooled	Percentile	84.7 [83.6, 85.6]	32.2 [31.0, 33.5]	49.0 [47.6, 50.4]
KUL3 / pooled	Welch	89.7 [88.8, 90.5]	30.8 [29.5, 32.1]	49.0 [47.6, 50.3]
KUL3 / reference	Percentile	86.3 [85.3, 87.2]	19.0 [18.0, 20.1]	47.5 [46.1, 48.8]
KUL3 / reference	Welch	88.1 [87.2, 89.0]	18.8 [17.7, 19.9]	48.2 [46.8, 49.6]

target changes. At five cells the direction reverses; shared randomness need not inflate inclusion at every sample size.

The increasing discrepancy at large tumor-arm counts has an analytic explanation. The estimate is $\hat{\theta} = f_N(\bar{Y}_T - \bar{Y}_N)$, whereas its error against the drawn benchmark is $\hat{\theta} - i_{\text{req}} = f_N(\bar{Y}_T - s\bar{Y}_N)$. For independent arm draws conditional on P , their error variances differ by

$$\text{Var}(\hat{\theta} - \theta_P \mid P) - \text{Var}(\hat{\theta} - i_{\text{req}} \mid P) = f_N^2(1 - s^2) \frac{\sigma_P^2}{n_N}, \quad (4)$$

where σ_P^2 is the source-pool expression variance and $n_N = 800$. At $s = 0.5$, the drawn benchmark removes three quarters of the reference-arm variance contribution from the error being assessed, while the bootstrap continues resampling both arms. This explains why near-perfect benchmark inclusion can accompany lower coverage of the generating parameter. Skewed finite-sample errors still determine the actual percentile-interval rates.

These precision runs assess four specified counts, without selecting a replacement threshold. The eight-seed grid below remains scored against its original benchmark and cannot be reinterpreted as fixed-pool calibration.

2.4 Welch as a control for bootstrap resampling noise

Welch intervals increase fixed-pool coverage at 50 cells by 1.5–5.0 percentage points on the same draws (paired Monte Carlo standard errors 0.23–0.32 points). They also reduce null rejection, but it remains 8.9–30.8% and every lower confidence endpoint exceeds 5%. Welch intervals have no bootstrap resampling noise, so their excess null rejection on the same draws shows that finite inner-bootstrap noise cannot be the sole explanation for the failure shared by both methods. Detection remains below 80% even at the upper confidence endpoints. Substituting Welch intervals therefore fails the joint requirements at 50 cells. Section 2.5 separately compares bootstrap budgets and skew-aware methods.

Why detection can fall as cell count rises. The apparent detection curve combines effect sensitivity with a changing false-positive rate. Figure 1 reproduces the pooled-draw trough with greater precision. For KUL3, percentile detection falls from 49.0% at 50 cells to 44.0% at 100, then rises to 57.7% at 800; null rejection falls from 32.2% to 20.3% to 6.5%. At 50 and 100 cells, the median sample-based standard error is respectively 0.66 and 0.86 times the exact pool-conditional standard deviation. The fraction of draws with that ratio below one half falls from 29.1% to 7.6%. These diagnostics support a loss of spuriously confident rejections as the sampled variance becomes better represented, before increasing signal-to-noise dominates. All-zero tumor samples occur in only 0.32% of the 50-cell draws and none at 100: the trough is not explained by all-zero samples alone.

The five-cell diagnostic makes the problem especially clear: pooled percentile null rejection is 66.1% in SMC and 71.9% in KUL3. Under the halving, 46.2% and 41.9% of tumor samples are all zero. Resampling cannot recover an unobserved positive tail, although the reference arm still contributes

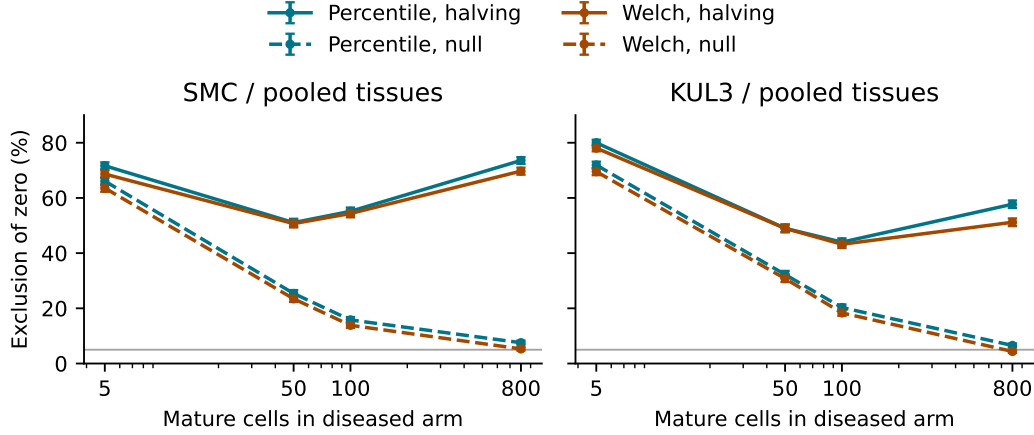


Figure 1: Detection must be read alongside null rejection. Pooled-tissue simulations use 5,000 studies per point; bars are 95% Wilson Monte Carlo intervals. Solid curves use a halving of expression and dashed curves the null. The horizontal line marks nominal 5% rejection. Five-cell intervals are diagnostic calculations before the reporting gate, which abstains below 20. Both methods use the same samples; their apparent sensitivity at sparse counts accompanies excessive false positives.

164 uncertainty. The original reporting gate already abstains here; the 50-cell failures concern returned
165 answers directly.

166 2.5 Testing the flag-wide band and skew-aware intervals

167 A separate fixed-design follow-up evaluates 30 and 50 cells, using 2,000 fresh studies per co-
168 hort/pool/shift: 32,000 studies and 160,000 interval evaluations. It compares percentile intervals
169 with 200 and 2,000 resamples, bias-corrected and accelerated (BCa) and bootstrap-*t* intervals with
170 2,000 resamples, and Welch. These skew-aware methods follow established bootstrap accuracy work
171 [DiCiccio and Efron, 1996]. All methods use the same study samples; bootstrap methods share
172 resamples. Coverage uses the fixed-pool target. Appendix A specifies implementation and numerical
173 checks.

174 At 30 cells, the original interval has 78.1–89.1% alternative coverage, 14.2–40.5% null rejection
175 and 48.5–67.9% detection. Flagging its width does not establish calibration. This evaluates one
176 interior point, not the whole 20–49-cell band. At 50 cells, increasing the percentile budget reduces
177 null rejection by 0.5–1.0 percentage points (paired Monte Carlo SEs 0.28–0.34 points), much less
178 than its excess over nominal.

179 The strongest null improvement occurs for bootstrap-*t* in SMC/reference: rejection falls from 8.45%
180 for percentile with 2,000 resamples to 6.05% [5.09, 7.18], a paired reduction of 2.40 points (SE
181 0.40). Alternative coverage is 92.6%, but detection falls from 76.3% to 65.9%. The improvement
182 does not transfer uniformly: in KUL3/pooled, BCa and bootstrap-*t* have 32.2% and 31.8% null
183 rejection, against 30.9% for the equal-budget percentile interval. All methods return intervals, so these
184 comparisons are not driven by abstention. No tested method meets the historical coverage/detection
185 requirements together with nominal null calibration at either count. This bounds the tested repair
186 attempt without selecting a new rule.

187 **Changing the resampling unit.** A separate comparison with two held-out patients also tests cluster
188 and paired patient-*t* intervals (Appendix A). At 50 cells, patient-*t* gives 2.77–6.29% null rejection
189 but only 4.21–8.02% detection among returned intervals; alternative abstention is 0.35–34.55%.
190 It equally weights shared patients, changing the cell-weighted target. Its saved coverage uses the
191 sampled-reference benchmark, so it cannot establish a fixed-pool repair. Full rates and abstentions
192 for both clustered methods are supplied; these results concern this design, not patient-level inference
193 generally.

Table 2: Fixed-pool evaluation inside the flag-wide band and at the reporting boundary. Each cohort/pool/count/shift uses 2,000 studies; all methods share samples. Bootstrap budgets follow method names. $C_{.5}$ is coverage at the halving, R_1 null rejection (one minus null coverage), and $D_{.5}$ detection at the halving, all in percent. Nominal coverage is 95%; the historical alternative-coverage minimum is 90% and detection requirement 80%. Maximum Monte Carlo SE is 1.12 percentage points; Wilson intervals and paired comparisons are supplied with the aggregate results. All intervals are finite.

Cohort / pool	Interval	30 cells: flag-wide			50 cells: report		
		$C_{.5}$	R_1	$D_{.5}$	$C_{.5}$	R_1	$D_{.5}$
SMC / pooled	Percentile, 200	80.5	31.0	53.6	87.8	22.8	48.9
	Percentile, 2000	82.1	30.9	52.6	88.4	22.2	48.0
	BCa, 2000	77.5	29.5	47.9	83.7	21.6	43.9
	Bootstrap- t , 2000	79.7	28.1	48.0	86.3	20.8	43.2
	Welch	84.3	30.5	53.1	90.1	22.0	48.3
SMC / reference	Percentile, 200	89.1	14.2	67.9	91.3	9.5	77.8
	Percentile, 2000	90.0	13.8	67.0	92.1	8.5	76.3
	BCa, 2000	89.8	11.8	57.8	92.0	7.8	68.9
	Bootstrap- t , 2000	91.2	9.0	53.8	92.6	6.1	65.9
	Welch	90.4	13.7	68.4	92.5	8.6	78.3
KUL3 / pooled	Percentile, 200	78.1	40.5	56.4	83.4	31.8	49.7
	Percentile, 2000	79.1	40.1	55.8	84.4	30.9	49.9
	BCa, 2000	72.6	40.3	54.4	77.5	32.2	48.4
	Bootstrap- t , 2000	75.3	40.4	55.2	79.9	31.8	49.7
	Welch	83.7	39.4	56.1	88.2	30.7	49.8
KUL3 / reference	Percentile, 200	78.8	26.3	48.5	86.3	18.7	47.7
	Percentile, 2000	79.4	25.7	47.9	87.3	18.2	46.6
	BCa, 2000	77.3	24.0	42.2	85.4	17.0	39.5
	Bootstrap- t , 2000	78.4	22.9	43.0	86.2	16.1	40.6
	Welch	80.5	26.3	49.7	88.8	18.6	48.9

Source shape and arm balance. A separate sensitivity study uses 2,000 studies and 2,000 percentile resamples per setting (Appendix A). At 50 tumor cells, empirical-pool null rejection is 9.70–31.80%; removing zeros and rescaling the positive counts to the original mean gives 7.10–12.65%, while a mean/variance-matched Gaussian gives 5.45%. Keeping the empirical pools but changing the reference count from 800 to 50 gives 7.45–11.45%. This implicates source shape and arm imbalance without isolating their individual contributions. Under the null, equal-sized independent arms have an exactly symmetric difference by exchangeability, even when the source is skewed. Balancing also changes variance and the reference sample size; alternative detection was not evaluated for this design change, so these rates do not justify discarding reference cells or establish a repair.

2.6 Count grids change candidate cutpoints

Historical grids returned reference-only binned candidates of 100, 90 and 70, but changed random streams, bin boundaries and replicate budgets together (Appendix B). To isolate these choices, we simulated the union of counts once per cohort, pool and each of eight seeds, with 200 outer replicates and 200 inner bootstrap draws. Every grid summary reuses the same outputs at shared counts. A *grid* is the set of mature-cell counts a sweep can evaluate, fixed by which mature-cell fractions it varies. The committed grid is the one recorded in the original design and reaches $\{0, 20, 40, 100, 200, 400, 800\}$; the filled and dense grids add support between and below those counts, and differ from it in nothing else. Fixed bins share boundaries across grids; adaptive bins recalculate them. Binned candidates are median counts within the selected bin, not estimated crossing locations.

Table 3 summarizes the candidates. The largest binomial Monte Carlo standard error in this grid is 0.035.

At the committed count of 50, SMC reference-only discrimination ranges from 75.5% to 81.0% across seeds, with both criteria met on two of eight seeds. KUL3 reference-only discrimination is only 45.0–55.0% at that count. The 200-replicate local crossings remain unresolved by the precision follow-up, which does not revisit counts 45, 60 or 70.

Table 3: Joint coverage-and-discrimination candidates from the controlled union-grid run. Entries give candidates with seed frequencies in parentheses where they vary; otherwise all eight seeds agree. Every grid summary reuses the same draws at shared counts. “None” means no tested count or bin qualifies.

cohort / pool	grid	fixed bins	adaptive bins	per count
SMC / reference	committed	100	100	100
	filled	70	65 (7), 90 (1)	50–80
	dense	70	70	45–80
KUL3 / reference	all three	400 (6), 800 (2)	same	same
Both / pooled	all three	None	None	None

A stable binned candidate of 70 is not a uniquely determined threshold: SMC’s first qualifying actual counts span 50–80 on the filled grid and 45–80 on the dense grid. Adding settings can change both a bin’s median count and its aggregate performance even when its boundaries stay fixed.

2.7 Patient influence and the cohort gap

Enumerating all 45 and 15 two-patient holdout pairs separates patient omission from repeated simulation. Keying streams to the sorted pair identifiers keeps every retained pair fixed. Raising the inner bootstrap budget from 200 to 1,000 and 5,000 leaves the first cohort’s candidate at 70, but moves the second’s 400 to 300 and back to 400. Omitting one source patient changes the joint coverage-and-discrimination candidate for none of ten patients in the first cohort and three of six in the second (KUL01, KUL30 and KUL31). For the coverage-only candidate, the corresponding counts are five of ten and zero of six. These checks quantify influence within each observed cohort. Appendix B gives the biological comparability limits.

The sensitivity windows cover counts 40, 50, 60, 70, 80, 90, 100, 110 in SMC and 200, 300, 400, 600, 800 in KUL3. These per-count summaries use a fixed pair schedule, so they are a separate experiment from the eight-seed union grid. Omitting KUL01 or KUL30 moves the joint candidate from 400 to 300; omitting KUL31 moves it to the lowest tested count, 200, with its lower boundary unobserved. The coverage-only baselines likewise occur at the lowest tested counts, 40 and 200. Their stability cannot locate a lower crossing outside those windows. Overlapping pairs do not add independent patients.

Expression dispersion partly explains the cohort gap. Holding tissue eligibility fixed at reference-only, the complete SMC and KUL3 mature-cell pools contain 662 and 844 cells. Across their 45 and 15 eligible patient pairs, median source-pool squared coefficients of variation ($c_P^2 = \sigma_P^2 / \mu_P^2$) are 4.35 and 16.99; median zero-expression fractions are 39.3% and 62.1%. The source means are 19.5 and 11.9 raw counts, respectively (medians over pools). The KUL3 contrast therefore has substantially more relative sampling noise despite its larger source-cell pool.

For the implemented binomial thinning, the tumor-cell variance is $s^2 \sigma_P^2 + s(1-s)\mu_P$. At $s = 0.5$, the exact conditional variance of the contrast relative to its squared signal is

$$\frac{\text{Var}(\hat{\theta} | P)}{\theta_P(0.5)^2} = \frac{c_P^2 + 1/\mu_P}{n_T} + \frac{4c_P^2}{n_N}. \quad (5)$$

With $n_T = 50$ and $n_N = 800$, the median signal-to-standard-deviation ratios are 3.02 in SMC and 1.53 in KUL3. This provides a distributional explanation for weaker KUL3 discrimination. It does not reproduce the exact 70-versus-400 candidate ratio: percentile-bootstrap behavior, pair heterogeneity and binning also affect that selection. The biological causes of the source-distribution difference remain unexplained.

Biological interpretation is a separate requirement. Interval calibration cannot establish a biological mechanism. The assumed retained-gene control, MS4A12, instead falls by -0.971 to -0.951 in non-sparse primary-cohort strata. Its depletion persists under broader cell inclusion and depth matching and appears in a separate bulk dataset. MS4A12 therefore does not serve as a retained-gene control in these data, a distinct failure of the panel assumption. These observations do not separate composition from within-cell change. Appendix E reports this control-panel audit as a limitation of the biological application.

3 A population-mismatch illustration

Changing the reference can also change the population being assessed. A constructed external-control example makes this distinction explicit. A fixed synthetic registry supplies a single-arm trial and three controls per enrollee, with enrollment favoring heavily pretreated strata and benefits varying by stratum and individual responsiveness. The intended finite-cohort target and the standardized estimator are

$$\tau_E = \frac{1}{|E|} \sum_{i \in E} \{Y_i(1) - Y_i(0)\}, \quad \hat{\tau}(w) = \sum_g w_g (\bar{Y}_{1g} - \bar{Y}_{0g}). \quad (6)$$

For enrolled records E and controls C , enrolled weights are $w_g = n_{Eg}/|E|$; pooled weights are $w_g = (n_{Eg} + n_{Cg})/(|E| + |C|)$. Deliberately using pooled weights in both estimator and observed-data reference makes their agreement an identity. It is a pedagogical construction, not an empirical frequency claim.

Table 4: Reference choice reverses the estimator ranking in the synthetic external-control example. Bias and RMSE use 200 studies with 1,600 enrolled records and 4,800 controls. Columns identify the population used to average observed stratum contrasts; only enrolled standardization matches the intended target. All entries are in abstract outcome units.

Validation reference	Pooled standardization		Enrolled standardization	
	Bias	RMSE	Bias	RMSE
Observed contrast, pooled	0.000	0.000	-0.872	0.876
Potential outcomes, pooled	+0.003	0.115	-0.869	0.875
Potential outcomes, enrolled	+0.885	0.892	+0.013	0.105

The informative control breaks that identity: the pooled potential-outcome reference still favors pooled standardization, whereas τ_E reverses the ranking (Table 4). Thus avoiding an identical reference is insufficient when it still targets the wrong population. The Monte Carlo study quantifies errors under this chosen case mix; it does not discover the algebraic equality. Under zero case-mix shift, pooled-standardization bias against τ_E is only -0.002 units despite the same identity. Under the full shift, weighting balances measured strata between arms but leaves a maximum standardized difference of 0.964 against the enrolled population. Between-arm balance does not establish population alignment. Appendix D specifies the 200-study-per-size design.

4 Implications for clinical-simulation validation

The audit measures a failure of the recorded reporting claim; the synthetic illustration explains why target-population alignment is a separate requirement. Three established checks make these distinctions operational.

State the target before constructing its reference. Distinguish a population treatment effect, a latent finite-cohort effect and an observed-data contrast. Name the population and the averaging weights. A reference need not be independent of the simulated dataset: τ_E is necessarily a property of the enrolled records. The requirement is that it represents the intended target, rather than merely reproducing the estimator.

Evaluate the interval and reporting policy jointly. Assess the whole decision rule using the interval that would actually be reported. Record bias and error against the chosen target, null rejection, detection at the specified effect, and abstention frequency. Report both conditional interval performance among returned answers and how often a valid answer is returned. A successful effect-detection rate can accompany excessive null rejection, so the former alone cannot validate an interval. For a nominal 95% interval, null rejection and null noncoverage are the same quantity; alternative coverage and detection answer different questions.

Separate rule selection from assessment and transfer. Separate selecting a rule from assessing it. If a scalar threshold is intended to guarantee performance above that threshold, qualification

294 must address the evaluated upper range rather than only its first passing point. Reserve independent
295 simulation draws for the final assessment, report rate uncertainty and the tested range, and distinguish
296 this from transfer to new patients or cohorts. Increasing resamples from an empirical pool improves
297 conditional numerical precision; it does not broaden that pool's biological support. For an external-
298 control application, the additional question is comparability across data sources, which requires direct
299 study-level validation such as the held-out comparisons of Ventz et al. [2019].

300 5 Conclusion

301 The audit separates three claims that had been conflated: recovering a reference value, covering the
302 intended parameter, and meeting the requirements of a reporting rule. Sharing the sampled reference
303 mean changes the error variance being assessed. Direct evaluation of the fixed-pool target finds
304 undercoverage at the halving and still worse coverage at the null. Increasing the bootstrap budget and
305 using skew-aware intervals improve some conditions but do not qualify either tested reporting count.
306 For this artifact, the practical action is to withdraw the claim that 50 cells is a validated reporting
307 boundary. Outputs can be retained as exploratory estimates with unvalidated uncertainty; a warning
308 about width is not a calibration guarantee. A replacement needs an explicit acceptable error rate,
309 adequate detection and independent assessment across its intended reporting range. New patient-level
310 evidence is needed before extending these conditional results to clinical decisions.

311 6 Responsible use, limitations and impact

312 **Misuse and limitations.** NOT ESTIMABLE must not be read as “no effect” or as evidence that
313 surviving cells are behaving normally. A separate within-cell analysis over three cohorts and 42
314 patients also remained unresolved because its comparability checks returned *undecided* (Appendix B).
315 The central evidence is a data-based simulation self-audit; neither it nor the constructed examples
316 validate a patient world model or a replacement reporting rule. The scope and conditional target are
317 stated in Section 1. Repeated cell draws and overlapping patient pairs do not measure uncertainty
318 over new patients. Poisson- and negative-binomial-based intervals remain untested; one point in the
319 flag-wide band does not qualify its full range. The draw-pool choice was originally underspecified.
320 The absence-defined labeling axis remains sensitive to sequencing depth (Appendix B). The external
321 search found no affirmative case, so the documented failure remains a self-audit rather than evidence
322 about its prevalence. The external-control example uses a synthetic registry and an illustrative
323 decision threshold; it does not establish clinical benefit or support a treatment decision.

324 **Data and impact.** The single-cell cohorts are public, de-identified and previously published. The
325 release policy covers derived summaries and code, rather than cell-level matrices. A system reporting
326 group-specific quantities should distinguish an unavailable comparison from an estimated null and
327 should evaluate the coverage and discrimination of any abstention rule. Publishing failed calibration
328 attempts helps prevent unsupported thresholds from being treated as validated.

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A Simulation settings

Experiment	Fixed settings and sampling units
Single-cell calibration	GSE132465 (SMC): 10 paired patients; GSE144735 (KUL3): 6. Target: GUCA2A; <code>stem_pole</code> axis, <code>lineage</code> resolution, <code>differentiated</code> mature label; reference (<code>normal</code>) weighting. Each replicate holds out two patients and draws 2,000 cells per synthetic arm from their eligible pool; reference mature fraction 0.40. Expression shifts are 1.0, 0.8, 0.5, 0.25; calibration uses 0.5.
Interval construction	A nominal 95% percentile interval from 200 resamples of mature cells within each synthetic arm, holding cell fractions fixed. This assesses the conditional intrinsic contrast; it does not estimate population uncertainty across patients. Coverage and discrimination condition on defined intervals. The historical inclusion score uses the drawn reference in Equation 2; the paired follow-up also evaluates fixed-pool coverage using Equation 3.

Precision follow-up and comparator. Five fresh seed streams (202609160–202609164), each with 1,000 studies per setting, provide 5,000 draws at each of four counts and two shifts in four cohort/pool combinations. Every draw uniformly chooses a two-patient pair, samples 800 reference and n_T tumor mature-cell counts independently with replacement, and thins tumor counts binomially. This is the original generator’s marginal law for this contrast, not a replay of its random stream. All 120 source-pair summaries match the earlier analysis. The vectorized percentile routine agrees with the original implementation on 32 actual-pool checks; the Welch implementation agrees with an independent library routine.

Writing $v_a = S_a^2/n_a$ for arm a , the Welch interval is $\hat{\theta} \pm t_{0.975,\nu} f_N \sqrt{v_N + v_T}$, where $\nu = (v_N + v_T)^2 / [v_N^2/(n_N - 1) + v_T^2/(n_T - 1)]$. Zero estimated total variance would give an unavailable comparator interval; none occurred. For the sparsity diagnostic, sample-based standard error is compared with $f_N \sqrt{\sigma_P^2/n_N + (s^2\sigma_P^2 + s(1-s)\mu_P)/n_T}$. Settings were fixed before this exploratory follow-up; aggregate exports retain widths, seed rates and paired differences.

Reporting-band and repair follow-up. Five fresh streams (202609170–202609174), 400 studies each, use the same pool-conditional sampling law at 30 and 50 cells. For BCa, bias correction uses midpoint ties; acceleration uses signed two-sample influence contributions $-(Y_{Ni} - \bar{Y}_N)/n_N$ and $(Y_{Ti} - \bar{Y}_T)/n_T$. Endpoints agree with SciPy’s independent two-sample jackknife implementation on continuous, discrete, sparse and zero-tumor samples. Bootstrap- t uses $t^* = (\hat{\theta}^* - \hat{\theta})/\widehat{SE}^*$ and reversed pivot quantiles, with Welch standard errors recomputed in each resample. Invalid BCa transforms or any undefined studentized pivot return an unavailable interval; none occurred. All 160,000 intervals are finite. The study settings were fixed before the run; this remains exploratory analysis on existing pools.

At least one adjusted BCa endpoint lies within the outermost ten bootstrap order positions in 5.2–56.3% of studies across settings. These finite-budget results therefore do not characterize BCa with arbitrarily precise tail quantiles. Seed rates, paired differences and numerical diagnostics accompany the aggregate summaries.

Source substitutions and clustered alternatives. The separately registered diagnosis crosses two counts (50, 800), four cohort/pool conditions, four source families and two arm-size designs. Each of its 64 rows uses five streams of 400 draws, 2,000 inner resamples and the same two-patient pools; all intervals are defined. The families are empirical counts, mean-rescaled positive counts, mean/skew-matched gamma (with some skew clipping), and mean/variance-matched Gaussian. All rates were reproduced from source data. Gaussian rows reuse standardized draws across cohorts. Substitution contrasts are not additive causal shares or lower bounds on a skew contribution. At 800 cells the designs coincide; empirical null rejection is 4.55–6.40%.

B Calibration details and limits

Historical grids. Before the controlled run, SMC reference-only binned candidates were 100, 90 and 70 on the committed, filled and dense grids, with 500, 500 and 200 replicates per setting. These

runs changed random streams and bin boundaries as well as evaluated counts; Section 2 separates those choices. KUL3 reference-only candidates were 400 or 800, and no pooled-draw candidate qualified. Its largest setting sampled 800 mature cells per arm with replacement from 844 distinct source cells, not additional patients.

Recovery accounting and unavailable contrasts. The four diagnostic sweeps comprise 1,711,200 replicates: 1,283,400 with nonzero requested intrinsic reference and 427,800 with zero requested reference, for which ratios are undefined. The first 50-replicate sweep has 119,600 draws, partitioned into 89,700 and 29,900, respectively. The maximum estimate–realized-reference residual is zero throughout. At a zero-cell setting the raw ratio is $1/(1 - s)$ when the denominator is nonzero. These raw diagnostic values arise from the harness setting an empty-arm mean to zero; they do not define an effect in absent cells. The reporting gate abstains and the interval is undefined. Separately, undefined intervals were previously counted as failures through comparisons with NaN; corrected rates exclude them and report abstentions. An all-abstaining bin has no coverage estimate. One historical seed gives $ok = wide = 45$, showing that a three-state specification need not yield three distinct fitted regions.

Biological scope and patient uncertainty. A separate comparability analysis over three cohorts and 42 patients remains undecided: patient-bootstrap $\log_2$ expression intervals cross the accepted $[-0.5, 0.5]$ boundaries in every cohort. Depth matching discards roughly three quarters of cells and pushes about half the remaining estimable rows below the count threshold. Within SMC, changing tissue eligibility changes the mature-cell pool from 662 to 3,824 cells; Section 2.7 holds tissue eligibility fixed.

C Supporting estimator and trial checks

Reference dependence. With nonempty stratum arms, saturated G-computation [Robins, 1986], inverse-probability weighting (IPW) using empirical stratum propensities, non-cross-fitted saturated augmented IPW (AIPW) and matching using every control all reproduce $T(D) = \sum_g (n_g/n)(\bar{Y}_{1g} - \bar{Y}_{0g})$. The IPW weights recover each stratum mean; the AIPW augmentation sums to zero within stratum. Ordinary least squares (OLS) with additive stratum indicators instead weights contrasts by normalized $n_g \hat{p}_g(1 - \hat{p}_g)$ [Angrist, 1998, Frisch and Waugh, 1933]. These known identities show why the reference determines what an exact recovery check establishes. Exact 1:1 allocation within strata makes OLS, standardized and unadjusted contrasts coincide; balance only in expectation does not guarantee this identity.

Learned-generator check. Seven generators comprise four Gaussian mixtures, a multilayer-perceptron (MLP) conditional Gaussian and two parametric controls. Training cohorts are simulated from a three-stratum model with confounded treatment assignment and a homogeneous additive effect of 3; learned generators are fitted to these cohorts, not to clinical data. Ten estimators are then applied to fresh synthetic cohorts from each fitted generator. Six sizes (100, 200, 500, 1000, 2000, 5000), six seed streams and 30 replicates schedule 75,600 evaluations on 7,560 cohorts; generator failures leave 75,580 evaluations on 7,558 cohorts. Of these, 74,726 estimate–drawn-reference pairs are finite. The 854 exclusions include 410 nonfinite references and 731 nonfinite estimates, overlapping in 287 evaluations.

Saturated G-computation has zero residual against $T(D)$ in all 42 generator–size settings because the equality holds on every evaluable synthetic dataset. It does not depend on learning the generator correctly. Nonsaturated ridge and MLP outcome fits can depart; the largest MLP residual under mixture generators is 3.3 outcome units. These checks concern the estimator–reference pair, not realism of learned patient trajectories.

D Synthetic external-control design

The fixed registry contains 50,000 simulated records with stratum shares (0.55, 0.30, 0.15), baseline outcomes $Y_0 \mid g \sim \mathcal{N}(\mu_g, 3^2)$, $\mu = (12, 8, 4)$, and effects $Y_1 - Y_0 = \delta_g r$, $\delta = (3, 1.5, 0.25)$, $r \sim \text{Gamma}(4, 0.25)$ (shape, scale). Responsiveness is fixed per record and hidden from estimators. Enrollment weights favor shares (0.15, 0.25, 0.60); both arms sample without replacement and are

disjoint. The pooled target is a mixture of enrolled and control records, not the registry population itself. Normal outcomes can be negative: this is an abstract continuous-outcome experiment, not a fitted progression-free-survival model.

There are 200 study draws at each of $n = 50, 100, 200, 400, 800, 1600$, using one seed (20260915) and one fixed registry: 1,200 studies, eight estimators and five references yield 48,000 comparisons: 47,969 finite values and 31 exclusions from one study with an empty stratum arm at $n = 50$. The registry uses the same seed. The intended finite-cohort truth averages both potential outcomes over enrolled records; it is available to the simulator, not to estimators observing one outcome per record. Five case-mix shifts and 50 balance draws at $n = 1600$ provide controls. Threshold crossings in the saved shift table compare medians, not replicate-level decisions.

The nine prespecified falsification checks did not fire. They verify the constructed comparison, not clinical realism or the frequency of the validation mistake.

E The retained-gene control

The control-panel audit compares saved unmatched and depth-matched single-cell summaries with a separate bulk dataset. Where both mature fractions equal one (all epithelium, without marker-based subdivision), MS4A12 still changes by -0.967 to -1.000 across the tested cohort/matching strata; CDX2 on the same cells changes by -0.396 to -0.293 . Depth-matched MS4A12 changes are -0.971 to -0.960 . These observations weaken a uniform labeling or depth artifact explanation; they do not eliminate every measurement artifact. Across nondegenerate strata, the median baseline is 2.56 CP10K and tumor expression 0.104 CP10K. Strata classified as degenerate have a median of one relevant tumor cell and cannot establish a precise complete loss.

In GSE39582 (585 bulk samples), the saved MS4A12 $\log_2$ fold change is -8.18 . This corroborates tissue-level depletion on a separate platform; it does not identify a within-cell mechanism. Lower-abundance genes retain more signal in the single-cell summaries, arguing against a simple universal abundance-floor explanation, not proving absence of gene-specific dropout. Repository chronology places panel freezing before dataset registration; it cannot establish what evidence was known outside the repository. The panel's prior expectation of retention therefore needs reassessment.

F External audit and research context

Search and classification. A pre-registered queue contained 41 entries; 19 were screened out and seven classified within 26 entries examined in order. A pre-inspection amendment relaxed unsuccessful queries without changing classification. All seven verdicts were NO: six repositories use generating-parameter or outside-sample references. The seventh compares cross-fitted partialling-out with full-sample OLS; their residual changes from 4.0×10^{-14} on noiseless linear data to 0.125 with noise, so agreement is restricted to the noiseless support. Classification had one reader, only that last repository was executed, and the sample is nonrandom. The virtual-control-arm query located no examinable code. These limits preclude a prevalence estimate.

Related scope. Cell-count filters [Crowell et al., 2020], composition analysis [Petukhov et al., 2022], selective prediction [Chow, 1970, El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017] and conformal coverage [Vovk et al., 2005, Angelopoulos and Bates, 2023] address related tasks; none automatically validates this intrinsic-effect interval or its reporting rule.

Documentary and implementation provenance. The anonymous record includes the original validation statement (Artifact A), interval criteria (Artifact B), and numerical context. Exact source mappings remain outside the review bundle. Saved tables carry commit hashes and seeds; these establish traceability, not statistical validity.

Use of large language models. Language-model assistants supported coding, figures, drafting and model audits under author direction. They helped identify defects and introduced some of those above. Authors remain responsible for all claims. Automated checks re-derive selected results and complement review; they do not verify every manuscript statement.